

VICTOR NGO

@ CONTACT

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📍 Murrieta, CA
🌐 <https://github.com/vngosb>

🎓 EDUCATION

Masters of Science
Bioinformatics
Johns Hopkins University
📅 2025
📍 Baltimore, MD
🎓 GPA: 3.6

Bachelors of Science
Biology
University of California, Riverside
📅 2021
📍 Riverside, CA
🎓 GPA: 3.5

★ TECHNICAL SKILLS

- **Diagnostics & Data Science**
CAP/CLIA, Root Cause Analysis, V&V, DHF Traceability, pandas, NumPy, ggplot2, PCA
- **Programming & Pipelines**
Python (Biopython), R, SQL (PostgreSQL), Bash, Nextflow, Docker, YAML
- **Genomics & Assay Design**
RNA-Seq, BLAST (In-silico Inclusivity/Exclusivity), Primer/Probe Analysis, Target Selection, GATK

📄 PROFESSIONAL SUMMARY

Bioinformatician and Data Analysis professional with a Master of Science in Bioinformatics and extensive experience supporting molecular diagnostic workflows within CAP/CLIA-regulated environments. Proven track record of developing reproducible NGS pipelines utilizing Python, R, Bash, and Nextflow. Adept at managing large-scale diagnostic datasets, executing root-cause analyses on assay performance, and collaborating across R&D, software, and laboratory teams to translate computational insights into high-performance diagnostic assays.

👤 WORK EXPERIENCE

QC Specialist II, Data Analysis **GENMARK DIAGNOSTICS (Roche)**

📅 2022 - current 📍 CARLSBAD, CA

- **Assay Performance & Root Cause Analysis:** Conducted statistical trend analysis on large-scale multiplex molecular diagnostic datasets to identify assay variability and out-of-specification (OOS) events, correlating diagnostic outputs with manufacturing variables to drive root-cause investigations.
- **Cross-Functional R&D Collaboration:** Partnered with R&D, Manufacturing, Software, and Data Analytics teams to troubleshoot sequence-level and system-level anomalies impacting assay sensitivity, specificity, and reliability.
- **V&V and Regulatory Traceability:** Executed validation and verification (V&V) activities for diagnostic assays within a CAP/CLIA-regulated environment, maintaining meticulous documentation and data integrity standards aligned with Design History Files (DHF)

❤️ PROJECTS

Germline Variant Calling Pipeline (In Progress)

- Implemented strict quality control and sequence filtering protocols to interpret genetic heterogeneity across public Illumina datasets.
- Developed an automated NGS alignment and variant identification workflow using BWA-MEM, SAMtools, and GATK to process whole exome sequencing (WES) data.

Nextflow RNA-Seq Pipeline (NASH vs Control)

- Engineered a reproducible, end-to-end RNA-seq pipeline utilizing Nextflow and Docker to optimize pipeline portability.
- Curated and processed public high-coverage transcriptomic datasets extracted from the NCBI Sequence Read Archive (SRA), utilizing Salmon and DESeq2 to identify candidate biomarkers and differential expression patterns.